Fertility and Housing Tenure Choice

Tommaso De Santo

08/07/2026

Since May

One-market lifecycle model of housing and fertility. What changed:

- **Sequential fertility**: birth-by-birth attempts under declining fecundity; children ever born and children at home tracked separately
- **Idiosyncratic income risk**: persistent after-tax earnings process, set externally
- **One housing market**: the two-location structure is set aside for the national question

Model

Environment

- One national housing market; stationary equilibrium with household population S
- Households live J four-year periods from age 18; retirement, then survival risk
- Two child states: n children ever born, m children currently at home
  - Children at home raise consumption and housing needs
  - Each child matures stochastically, leaves the household, and enters the economy as a young adult
- Idiosyncratic persistent earnings risk; wages and the interest rate are exogenous
- Households rent, or own a discrete house size on a ladder financed with a mortgage
- Government: property tax rebated lump sum; payroll tax funds a pension
- New households enter as young renters: matured children plus an outside inflow

Preferences

Flow utility over consumption and housing services:

$$u(c, \mathbf{h}; m) = \frac{1}{1-\sigma}\left[\frac{c^{\alpha}\left(\mathbf{h} - \underline{h}\, m\right)^{1-\alpha}}{e(m)}\right]^{1-\sigma} + \psi_{\text{child}}\, m.$$

- **Space requirement.** Each child at home absorbs $\underline{h}$ rooms before housing yields services
- **Equivalence scale.** $e(0) = 1$ and $e' > 0$: a larger family needs more resources to reach the same utility
- **Owner premium.** $\mathbf{h} = \chi h$ for owners, with $\chi > 1$; $\mathbf{h} = h^R$ for renters

Bequests

Warm-glow utility at death over terminal wealth:

$$\mathcal{B}(\widetilde{b}) = \theta_0 \frac{(\theta_1 + \widetilde{b})^{1-\sigma}}{1-\sigma}, \qquad \widetilde{b} = b + ph.$$

- θ_0 governs the strength of the bequest motive; θ_1 makes bequests a luxury
- Estates are valued by parents but transfer to no one: no intergenerational wealth flow

Fertility and Child Maturation

Between ages 18 and 45, households decide each period whether to try for a child — a discrete choice with taste shocks, with separate scales κ_f before and after the first child.

Trying yields a birth with the conception probability

$$\pi(\mathrm{age}) = 1 - \omega_1 \, e^{\omega_2(\mathrm{age}-18)}, \qquad \pi = 0 \text{ after } 45,$$

which falls from 0.98 at 18 to 0.62 at 40: postponed births are increasingly likely never to happen.

Each child lives at home for 18 years on average (stochastic maturation), raising the equivalence scale and the space requirement; matured children enter the economy as young households.

Earnings and Institutions

Idiosyncratic persistent earnings risk over the lifecycle:

$$\log y_{j+1} = \rho \log y_j + \varepsilon_{j+1}, \qquad \varepsilon \sim N(0, \sigma_\varepsilon^2),$$

discretized and combined with an age profile of earnings.

- Earnings are after tax: a progressive retention function applied to gross earnings
- Retirees receive a stationary pension funded by a payroll tax
- Property tax on owners, rebated as a lump-sum transfer

Housing, Tenure, and Budget Constraints

Households rent a continuous quantity up to a cap, or own a discrete size on a ladder:

$$h^R \in (0, \bar{h}^R], \qquad h \in \{H_1, \ldots, H_K\}.$$

Tenure segmentation: $\max H_k > \bar{h}^R$, so family-sized space is reachable only through ownership.

Renters

$$c + b' + r\,h^R = (1+q)\,b + y(j), \qquad b' \geq 0.$$

Owners, with prior stock h_{-1}

$$c + b' + (\delta + \tau_H)\,ph + \mathbf{1}\{h \neq h_{-1}\}\big[ph - (1-\psi)\,ph_{-1}\big] = (1+q)\,b + y(j), \qquad b' \geq -\phi\,ph.$$

- q: interest rate; δ: maintenance; τ_H: property tax; ψ: transaction cost; $1-\phi$: down-payment share

Housing Supply and Market Clearing

Housing supply slopes upward in levels, and rents map into prices through user cost:

$$H^S = H_0 \left(\frac{r}{\bar{r}}\right)^{\xi}, \qquad r = (q + \delta + \tau_H)\, p.$$

- One market-clearing condition: aggregate housing services demanded equal H^S
- Supply is in levels, not per household: a permanently larger population permanently raises rents
- Stationary equilibrium: the price, the transfer, and the distribution are constant, and the government budget clears

The Household's Problem

Within the period, a household first decides whether to try for a child, then chooses tenure, housing, and saving:

$$V_j(b, h, y, n, m) = \mathbb{E}_\varepsilon \max_{a \in \{0,1\}} \Big\{ W_j^a(b, h, y, n, m) + \varepsilon_a \Big\},$$

$$W_j^a = \max_{\text{rent/own},\, h',\, h^{R\prime},\, b'} \; u(c, \mathbf{h}; m) + \beta\, \mathbb{E}\big[V_{j+1}(b', h', y', n', m') \big]$$

subject to the renter or owner budget and the collateral constraint; a successful try ($a = 1$, probability π_j) adds a child next period.

In old age there are no fertility decisions; survival risk and the warm glow at death close the problem.

Calibration

Externally Calibrated Parameters

	Object	Value	Source
Pref.	σ risk aversion	2.0	standard
Fin.	q interest (annual)	0.04	standard
	δ depreciation (annual)	0.02	Greaney et al.
	τ_H property tax (annual)	0.01	US average
	ϕ financed share	0.80	20% down payment
	ψ transaction cost	0.06	Greaney et al.
Life	17 four-year periods	ages 18–85	retirement 65
Income	ρ, σ_ε, HSV τ	0.914, 0.169, 0.181	FL $\times$ HSV
	τ_{pay} payroll	0.179	Greaney et al.
Children	maturation hazard	2/9 per period	18 expected years
	fecundity ω_1, ω_2	0.02, 0.134	conception schedule
Housing	ladder, rental cap	$\{2, \ldots, 11\}$, 6 rooms	room units

External restriction on the search: annual $\beta \geq 0.94$.

Internally Calibrated Parameters

Param	Description	Estimate	Bounds	Near bound?
β (annual)	Discount factor	0.9946	$[0.94, 0.9995]$	no
ψ_{child}	Child flow utility	0.000	$[0, 3]$	**at lower**
κ_f	Attempt taste scale, first birth	2.987	$[0.02, 50]$	no
κ_f^{cont}	Attempt taste scale, later births	1.713	$[0.02, 50]$	no
χ	Owner service premium	1.037	$[0.1, 5]$	no
H_0	Housing supply intercept	9.873	$[0.2, 80]$	no
θ_0	Bequest strength	0.159	$[0, 8]$	**near lower**
θ_1	Bequest shifter	0.094	$[0.02, 16]$	**near lower**
$\underline{h}$	Child room floor	0.614	$[0.1, 1.8]$	no

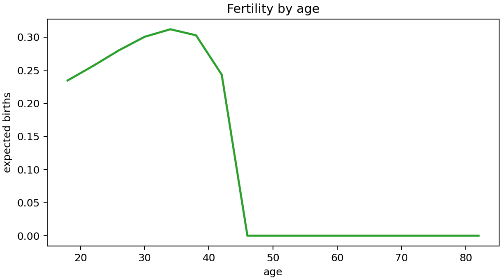
Nine parameters, twelve moments. ψ_{child} is estimated at zero, and the bequest parameters sit near their lower bounds.

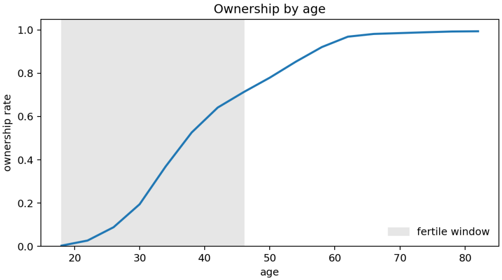
Model Fit

	Moment	Target	Model	Gap	Contr.
Fertility	Completed fertility	1.918	1.967	+0.049	3.4
	Childlessness	0.188	0.104	−0.084	**121.5**
	Mean age at first birth	25.31	25.57	+0.26	1.1
	First births at 30+	0.270	0.327	+0.057	**51.5**
Housing	Rooms response to first birth	0.664	0.690	+0.026	0.6
	Rooms, 3+ vs. 1–2 children	0.368	0.359	−0.009	0.2
	Family ownership gap	0.168	0.162	−0.006	0.5
	Ownership rate	0.575	0.582	+0.007	0.1
	Aggregate mean rooms	5.780	6.478	+0.698	5.8
Wealth	Wealth / earnings	6.873	5.794	−1.079	7.3
	Bequest flow / wealth	0.0088	0.0087	−0.0001	0.0
	Old wealth p90/p50	3.448	2.090	−1.358	**105.1**

The remaining misses are childlessness, the dispersion of old-age wealth, and the share of late first births; additional search does not improve them.

Lifecycle Profiles





Population Closure

The Population Question

Completed fertility is below replacement: on its own, the household population shrinks. What sustains a stationary economy — in the model as in the U.S. — is immigration.

This is an economic question before it is a computational one. When policy raises births, what gives?

- If immigration is unchanged, the population grows — and presses on a housing supply that does not scale with it
- If instead the population is held fixed, fewer immigrants are needed
- Letting entry respond to the value of living here would require an elasticity that nothing in the data identifies

U.S. immigration is rationed and oversubscribed: how many enter is set by the quota, not by the marginal entrant's indifference. The closure adopted here follows: the inflow is a policy quantity, fixed across counterfactuals.

Population Balance

Entrants each period are retained matured children plus the immigration inflow M:

$$\underbrace{S\,E_0}_{\text{entrants needed}} = \underbrace{R\,S\,B_0}_{\text{retained matured children}} + M,$$

with E_0 the replacement flow per household, B_0 the matured-children flow, and R the share who stay.

Given the fixed inflow, the stationary population

$$S = \frac{M}{E_0 - R\,B_0}$$

exists and is stable: as the population shrinks, a fixed inflow becomes a growing share, and the decline stops.

R and M are pinned at the baseline by two observables: the outside-origin share of young entrants and the population normalization.

Population in Counterfactuals

With R and M fixed, population responds to policy through births alone. Differentiating the balance condition:

$$\%\Delta S = \frac{1 - s^{out}}{s^{out}} \%\Delta B_0,$$

where s^{out} is the immigrant share of entrants. Each immigrant household founds a family line that fades out at the native replacement rate; a policy that raises births makes every line last longer.

At the measured share (16.9% of young entrants, ACS), the multiplier is about five. Housing pushes the other way: a larger population raises rents, which discourages births.

Policy Results

Funded Property-Tax and Grant Reforms

Raise the property tax from 1% to 2%; rebate lump-sum, or fund a purchase grant for family-sized homes. All changes vs. the funded 1% baseline, percent:

Arm	Policy	TFR	Births	Househ.	$-\Delta$Imm.	Price	Rent
Floor	2% tax	+0.38	+0.33	+1.70	1.74	−17.5	−6.1
Floor	tax + grant	+0.75	+0.67	+3.46	3.52	−17.3	−5.8
Tilt	2% tax	+0.20	+0.18	+0.90	0.92	−17.7	−6.3
Tilt	tax + grant	+0.49	+0.44	+2.23	2.26	−17.2	−5.7

In U.S. units, +0.33–0.67% of births is roughly 12–24 thousand additional babies per year. The tax lowers the asset price, so the down payment falls by 17.5%, while renter rents — which include the tax — fall by 6%.

Reading the Population Numbers

The same number can be read three ways:

- more households in the long run, at a fixed immigration inflow;
- fewer immigrants required to hold the population fixed;
- the share of the fertility gap closed by the policy.

Housing feedback is present but small: the demographic arithmetic overstates the response by at most a tenth of a percentage point.

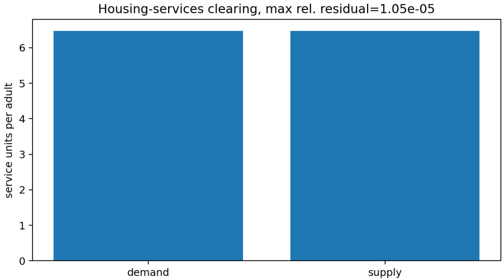
Timing: the birth and immigration flows move immediately, while the household count converges over generations — half of the adjustment takes roughly a century.

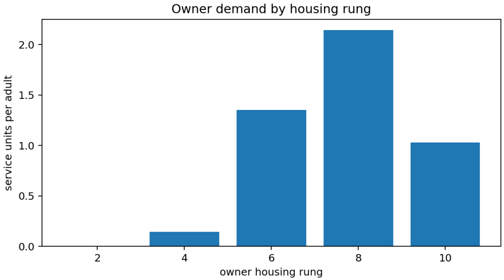
Open Decisions

1. **Anchoring the inflow.** Among household heads aged 18–22, recent immigrants are only 3.7% — but immigrants mostly arrive older than the model's entry age. Either anchor the inflow to total net migration in entrant equivalents (roughly 17–22%, close to the 16.9% in use), or model entry at several ages
2. **The fertility block.** The closure requires natives to replace at least 83% of each cohort; the current fit sits close to that edge, and the childlessness and old-age wealth misses are limitations of the mechanism rather than of the search
3. **Welfare.** With different populations across counterfactuals, welfare is compared per household

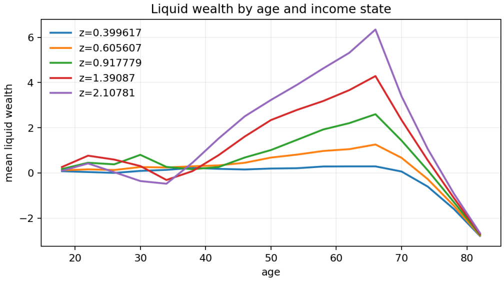
Appendix

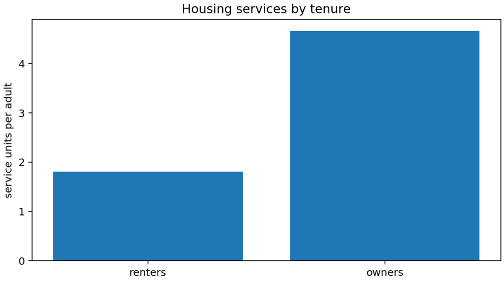
Appendix: Housing Market Diagnostics



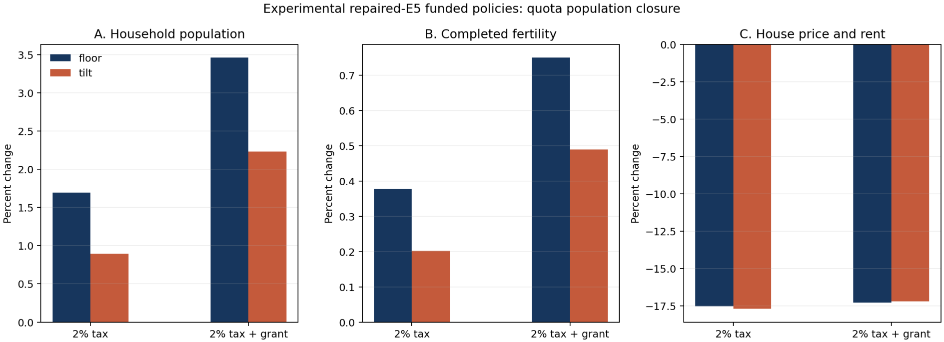


Appendix: Wealth and Tenure Diagnostics

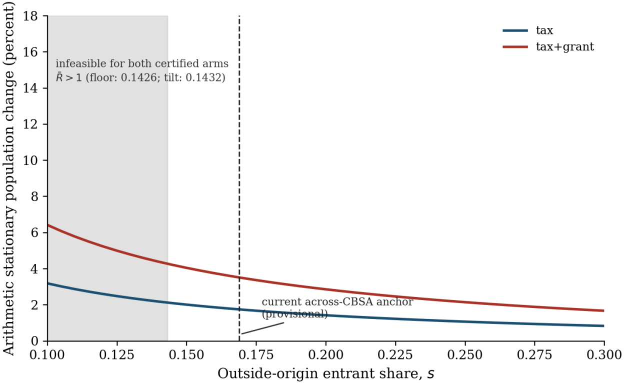




Appendix: Policy Comparison



Appendix: Anchor Sensitivity and Convergence Speed



Stationary population response as a function of the outside-origin share s^{out}, with the infeasible region ($s^{out} < 0.143$) shaded. Convergence: generational decay $\rho = \bar{R}B_0/E_0 = 0.831$ gives a half-life of 97–112 years ($g = 26$–30); an age-structured variant with uniform birth timing gives ~ 230 years (upper bound).

Appendix: The Retired Logit as a Sensitivity Row

The old protocol's numbers survive as a labeled robustness row, not a model:

Arm	Policy	Quota population	Logit sensitivity ($\lambda = 2$, unidentified)
Floor	2% tax	$+1.70$	$+9.98$
Floor	tax + grant	$+3.46$	$+10.10$
Tilt	2% tax	$+0.90$	$+8.99$
Tilt	tax + grant	$+2.23$	$+9.06$

The gap between columns is what an unidentified taste parameter was contributing. The logit row bounds "what if entry responded to values" without pretending to identify it.

Appendix: National Entry-Anchor Measurement (ACS 2012–2023)

Measure (household heads unless noted)	Share	Multiplier $(1-s)/s$
Ages 18–22, foreign-born, arrived $\leq$ 4y (headline)	3.71%	25.9
Ages 18–22, foreign-born, arrived $\leq$ 8y	5.36%	17.7
Ages 18–25, foreign-born, arrived $\leq$ 4y	3.47%	27.9
Ages 18–22, foreign-born stock (upper bound)	9.04%	10.1
Persons 18–22, foreign-born, arrived $\leq$ 4y	2.89%	33.7
Quota feasibility floor	14.3%	—
Age-aggregated net-migration bracket (to verify)	17–22%	3.5–4.9

Every entry-age-window measure is infeasible; the age-aggregated concept is not. The two are different objects because immigrants mostly arrive above the stylized entry age — the anchor decision on the Open Decisions slide.

Appendix: Multi-Start Record (overnight)

16 fresh chains, unchanged 12-moment / 9-parameter contract, 225-minute budget each, strict-repeat certification:

	Best	Median	Worst
Fresh chains (seeds 09–24)	263.5	$\sim$284	370.5
Certified incumbent (Aug 5, chain 6)	249.186 — **stands**		

All 16 eligible with exact strict repeats on the run's best; market residuals $\leq 2.2 \times 10^{-5}$. With 24 chains total across the two arrays, the remaining misses are search-converged.